Speech Recognition (DSAI 456)

Lecture 3: Spectral Front End

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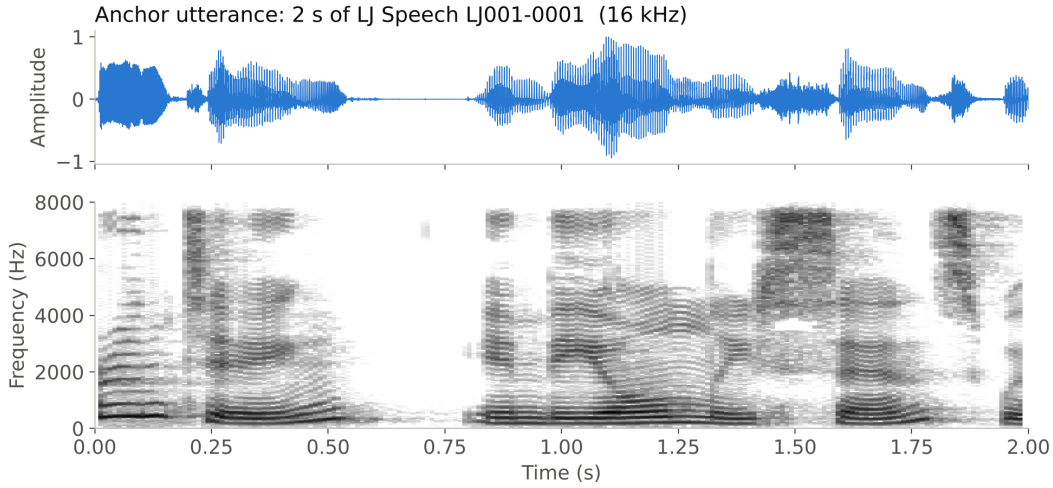
Agenda

Objectives: by the end of this lecture you should be able to compute DFT bin frequencies and resolution, explain why we window frames and what spectral leakage is, describe what the STFT and the spectrogram contain (and their shapes), build a log-mel spectrogram step by step, and say why each step of the MFCC pipeline exists.

- **The frequency domain**: DFT and FFT
- **Frames and windows**: framing, leakage, Hamming and Hann
- **STFT and spectrogram**: the time-frequency trade-off
- **Mel and log**: mel scale, filterbank, dB vs. mel, log-mel
- **MFCC**: reading homework
- **The whole pipeline**: waveform to features

Recap: Last Week

- Speech is 16,000 numbers per second. We need to cut it into frames because it is non-stationary.
- A **spectrogram** shows formants, bursts, and silence in one picture.
- Today: how is that picture *computed*, and what do we feed a model instead of raw samples?



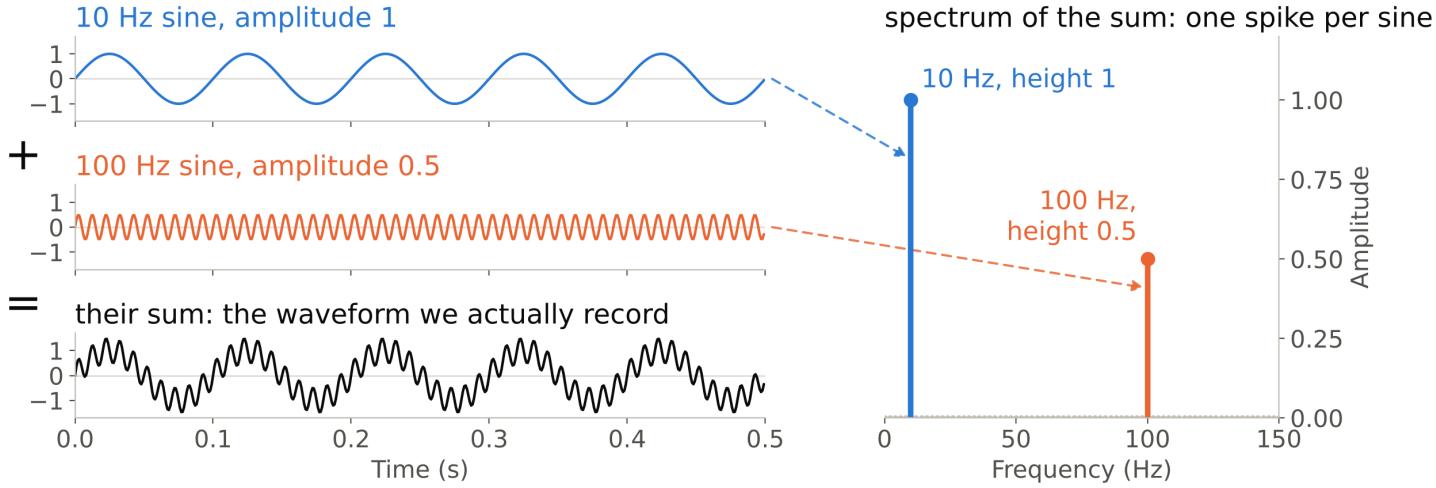
Part 1

The Frequency Domain

Fourier Transform

Any Wave Is a Sum of Sines

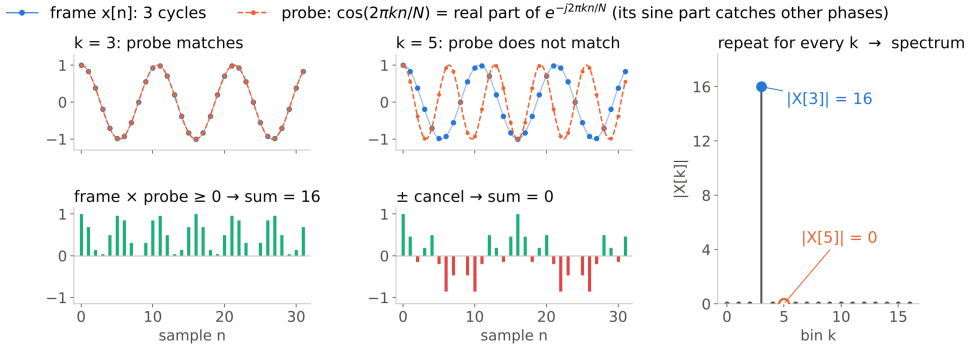
- **Fourier's insight**: every signal can be written as a sum of sine waves with different frequencies, amplitudes, and phases.
- The **spectrum** lists which frequencies are present and how strong each one is.
- A spectrum is another *representation* of the same signal.



Discrete Fourier Transform (DFT)

$$X[k] = \sum_{n=0}^{N-1} \underbrace{x[n]}_{\text{frame of } N \text{ samples}} \cdot \underbrace{e^{-j2\pi kn/N}}_{\text{probe wave: } k \text{ cycles per frame}}$$

- **Read it as**: multiply frame and probe sample by sample, then add up: a **large** sum if the frame contains that frequency; otherwise they cancel.
- $|X[k]|$ = **how much** of that frequency is present.



Fast Fourier Transform (FFT)

- The **FFT** computes exactly the same DFT in $O(N \log N)$ instead of $O(N^2)$ operations.
- The (radix-2) FFT needs N to be a power of 2.
  - If $L = 400$ samples per frame then $N = 2^9 = 512$ (Whisper uses $N = 400$).
  - We **zero-pad** with 112 zeros.

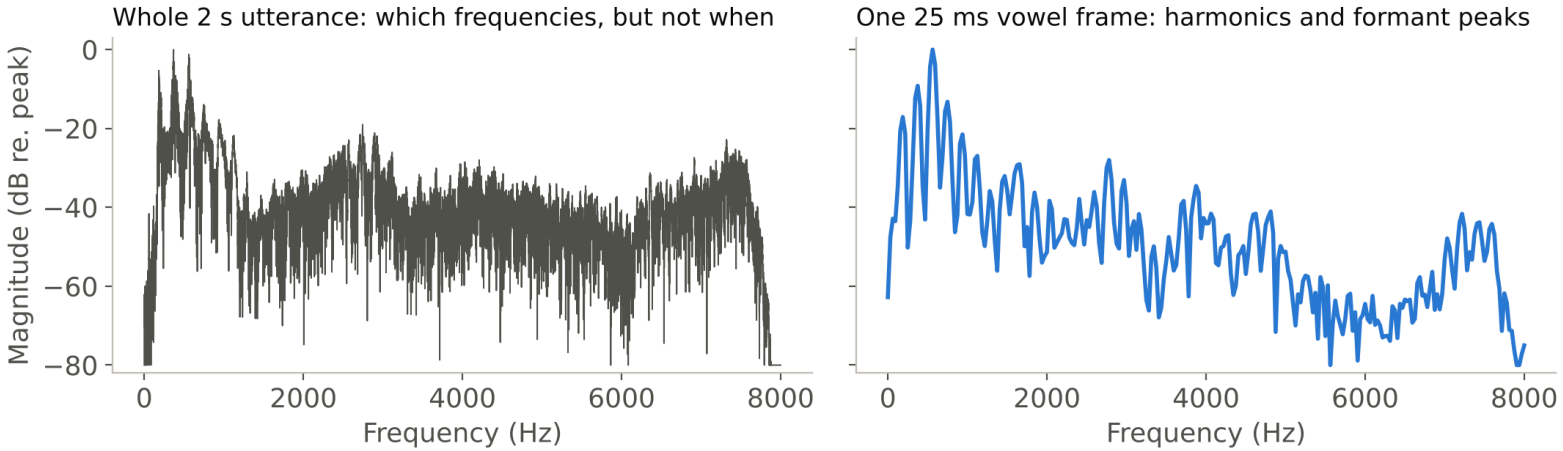
Part 2

Frames and Windows

Framing, leakage, Hamming and Hann

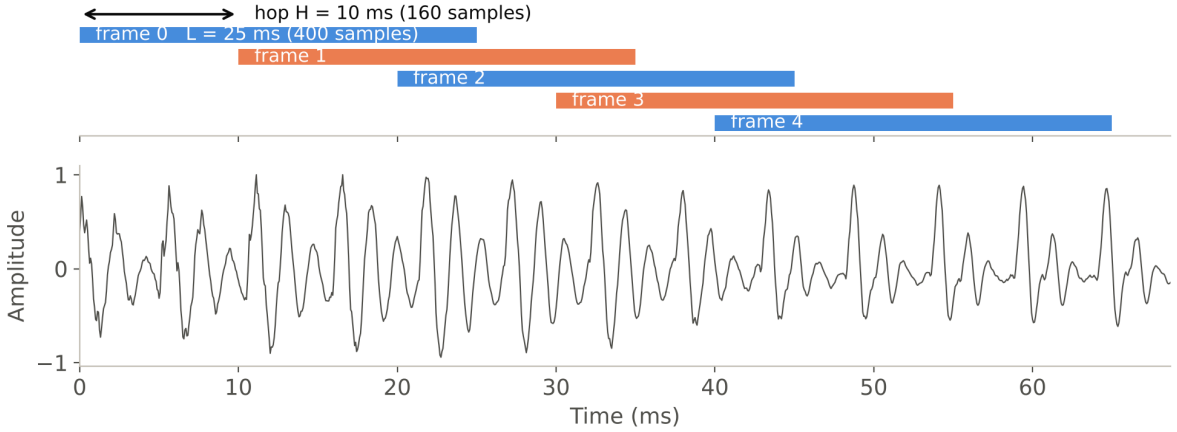
Why One DFT of the Utterance Is Not Enough

- The *magnitude* spectrum of the whole 2 s utterance says *which* frequencies occurred, but not *when*: the timing is hidden in the phase, which we discard, so every vowel is mixed together.
- A DFT of one short vowel frame shows its harmonics and the formant peaks that identify the phone.
- So we compute **one spectrum per short frame**, where the signal is roughly stationary.



Framing: Frame Length and Hop

- **Frame length** L (window size): 25 ms. **Hop** H (stride): 10 ms, so consecutive frames overlap by 15 ms.
- At $f_s = 16$ kHz: $L = 400$ samples and $H = 160$ samples.
- A signal of N_{tot} samples gives $T = 1 + \lfloor (N_{tot} - L)/H \rfloor$ frames.
- Our 2 s anchor is 32,000 samples, so $T = 1 + \lfloor 31600/160 \rfloor = 198$ frames.



Windowing: Cutting Out a Frame

- Extracting a frame means multiplying the signal $s[n]$ by a **window** $w[n]$ that is zero outside the frame:

$$y[n] = w[n]\, s[n]$$

- The simplest window is **rectangular**: keep the samples as they are inside the frame, zero elsewhere.

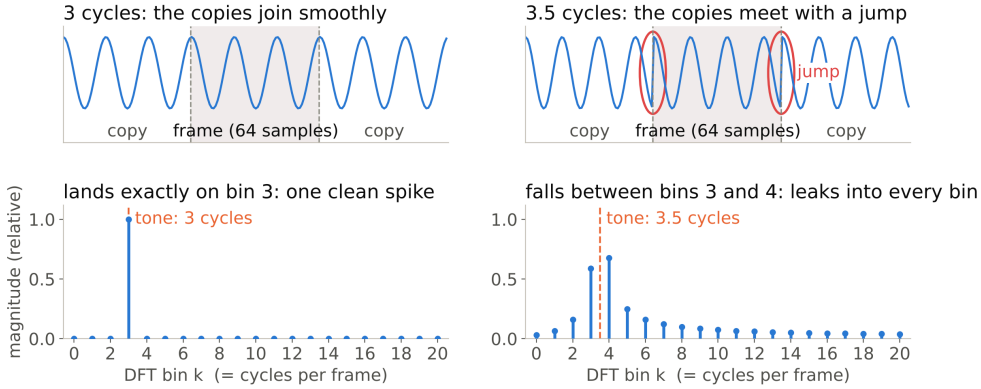
$$w_{\text{rect}}[n] = \begin{cases} 1 & 0 \le n \le L-1 \\ 0 & \text{otherwise} \end{cases}$$

- Cutting a frame with a rectangular window is what we implicitly did so far. It has a hidden cost.

Important
The DFT treats the frame as one period of a signal that repeats forever. If the two edges do not match, the repeated signal has a jump.

Spectral Leakage

- DFT bins sit at **whole numbers of cycles per frame**: bin k = exactly k cycles.
- **3 cycles** (left) lands on bin 3: the copies join smoothly → **one clean spike**.
- **3.5 cycles** (right) falls between bins 3 and 4: the copies meet with a **jump**, which needs many frequencies to describe → the energy **leaks** into every bin.

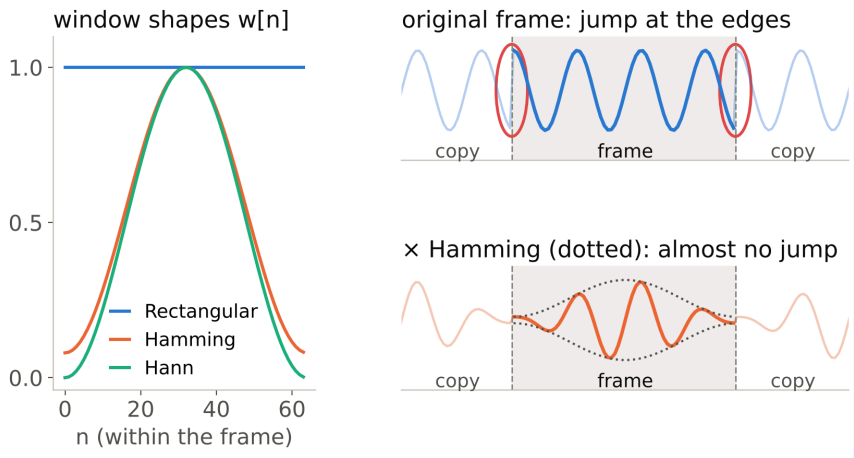


Window Shapes: Hamming and Hann

Taper the frame toward zero at both edges, so the copies meet with (almost) no jump:

$$w_{\text{Hamming}}[n] = 0.54 - 0.46\cos\left(\frac{2\pi n}{L}\right), \qquad w_{\text{Hann}}[n] = 0.5 - 0.5\cos\left(\frac{2\pi n}{L}\right)$$

for $0 \le n \le L-1$ (the periodic form, as in librosa; some toolkits divide by $L-1$).



- The price: the main lobe gets **wider** (slightly blurrier frequency), but the side lobes drop a lot.

Part 3

STFT and the Spectrogram

Stacking spectra over time

Short-Time Fourier Transform (STFT)

Put framing, windowing, and the DFT together. For frame index m and bin k:

$$X[m, k] = \sum_{n=0}^{L-1} w[n]\, x[mH + n]\, e^{-j2\pi kn/N}$$

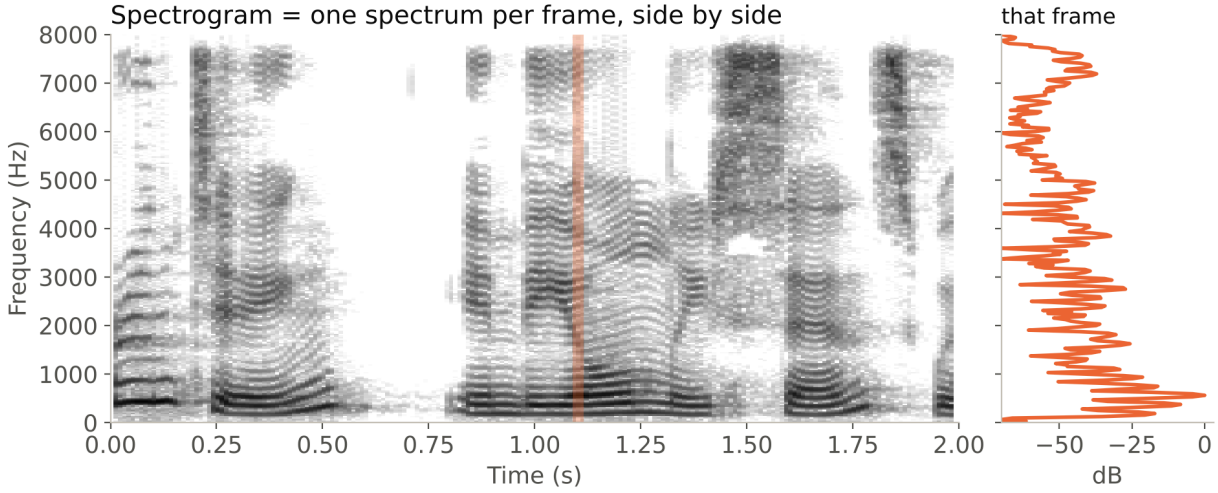
1. Take the L samples starting at mH.
2. Multiply them by the window $w[n]$.
3. Zero-pad to N samples and take the DFT.
4. Move forward by H samples and repeat.

The result is a complex matrix with one row per frame and one column per bin.

> **Important**
> Frame length *L* and hop *H* are the same ones from last week, now with a formula behind them.

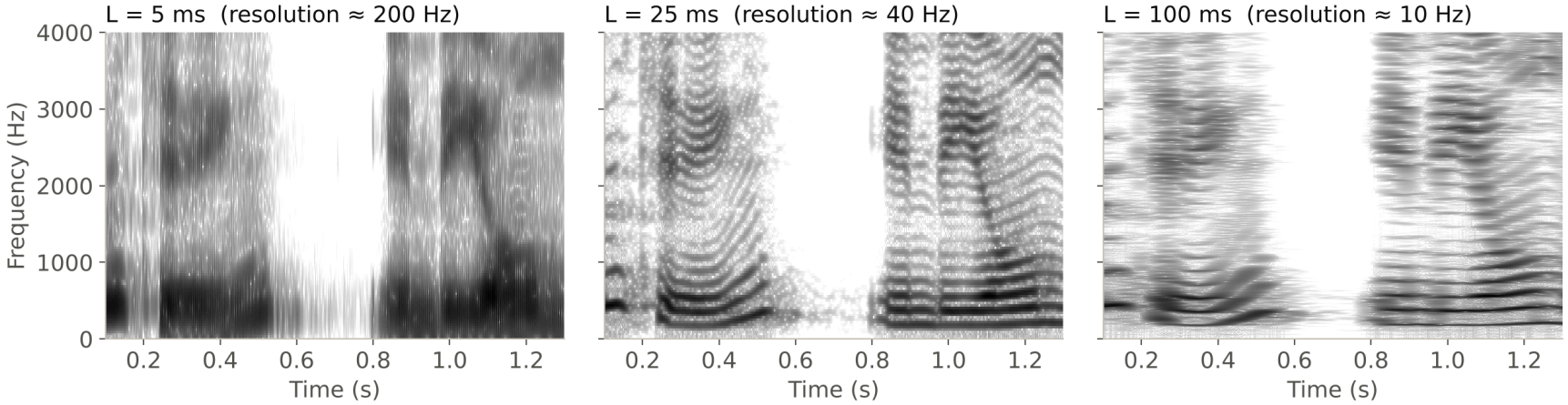
From STFT to Spectrogram

- Keep the magnitude and put it on a decibel scale, $S[m,k] = 20\log_{10}|X[m,k]|$, so quiet parts stay visible (why the log helps: Part 4).
- A spectrogram is **one spectrum per frame, placed side by side**.
- Shape: $T \times (N/2+1)$.



The Time–Frequency Trade-off

- **You cannot have both**: a short window is sharp in time and blurry in frequency, a long window the opposite.



Important
Same signal, same hop. Only the window length changed: 5 ms, 25 ms, 100 ms.

Part 4

The Mel Filterbank and the Log

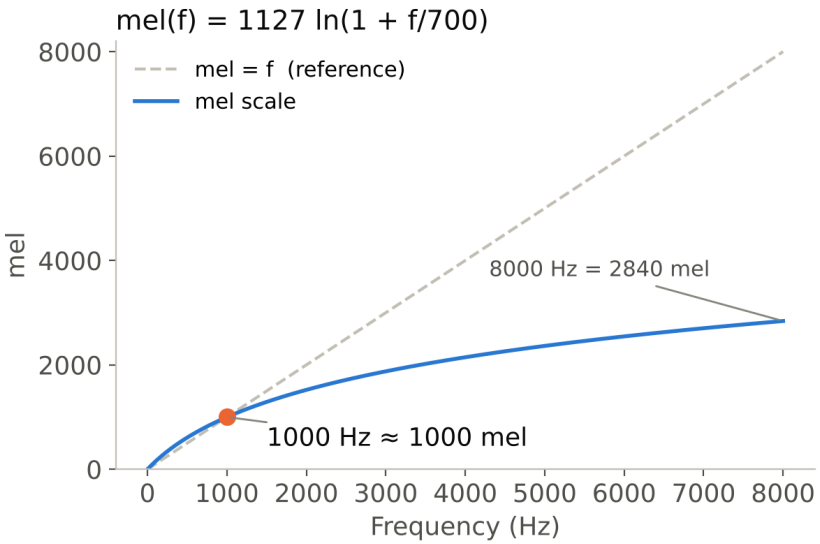
From 257 linear bins to 80 perceptual channels

Why Mel? The Mel Scale

- Pitch perception is roughly **linear below 1 kHz and logarithmic above**: 200 → 300 Hz is a big step (119 mel), 6200 → 6300 Hz a small one (16 mel).
- The **mel scale** puts equal perceived pitch steps at equal distances:

$$\mathrm{mel}(f) = 1127 \ln\left(1 + \frac{f}{700}\right)$$

- 1000 Hz ≈ 1000 mel; 8000 Hz ≈ 2840 mel.

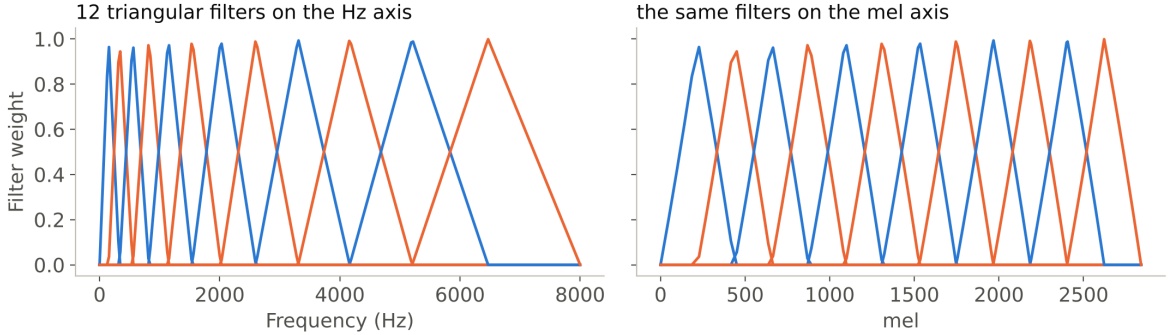


The Mel Filterbank

1. Choose $M = 12$ channels. Place $M + 2$ points **equally spaced in mel** from 0 to 8000 Hz (0 to 2840 mel), then convert back: $f = 700\,(e^{\,\mathrm{mel}/1127} - 1)$.

2. Filter m is a **triangle** from point m to point $m + 2$, with weight 1 at point $m + 1$.

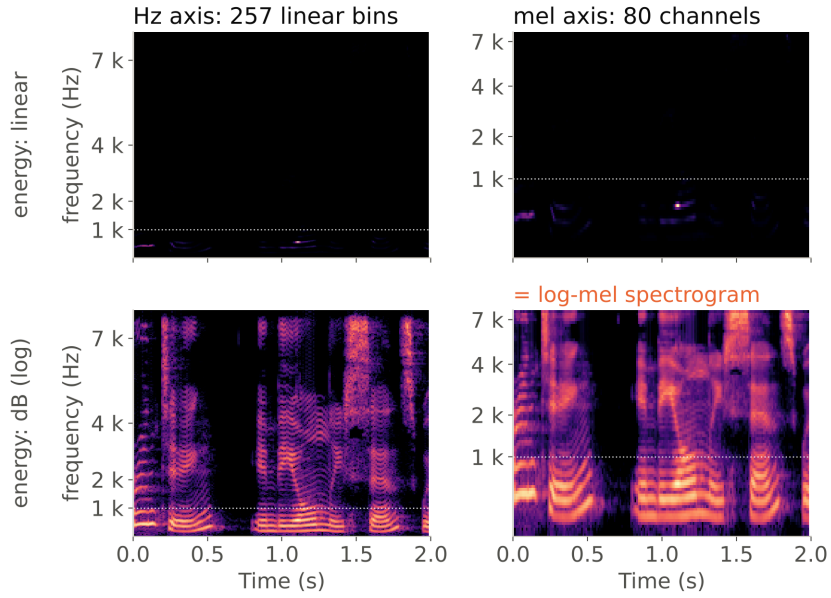
3. Each channel adds up the power under its triangle: $E_m[t] = \sum_k H_m[k]\,|X[t,k]|^2$. For all frames at once this is one matrix product: $(198 \times 257) \cdot (257 \times 12) \to 198 \times 12$.



Note
Equal width in mel = narrow filters at low Hz, wide ones at high Hz.

The Log Scales

- Two log-like scales on **two different axes**:
  - **mel** warps the **frequency** axis (pitch);
  - **dB / log** compresses the **energy** axis (loudness).
- Log step: $\log\left(E_m[t] + \epsilon\right)$, or $10 \log_{10} E_m[t]$ in dB (ϵ avoids $\log 0$).
- **Dynamic range**: a vowel can carry ~30 dB (1000×) more power than a weak fricative; on a linear scale only the loudest spots show (top row).



Log-Mel: The Standard Neural Input

`waveform → frame → window → |DFT|² → mel filterbank → log`

- The output is a **log-mel spectrogram**: one vector of 80 (or 128) channels per 10 ms frame.
- Before the network, features are usually **normalized**: zero mean and unit variance (per utterance or over the training set), or a fixed rescaling to about $[-1, 1]$ (Whisper).
- The encoders in Weeks 5–9 (CTC models, Conformer, wav2vec 2.0 and later) start from this representation or learn something similar directly from the waveform.

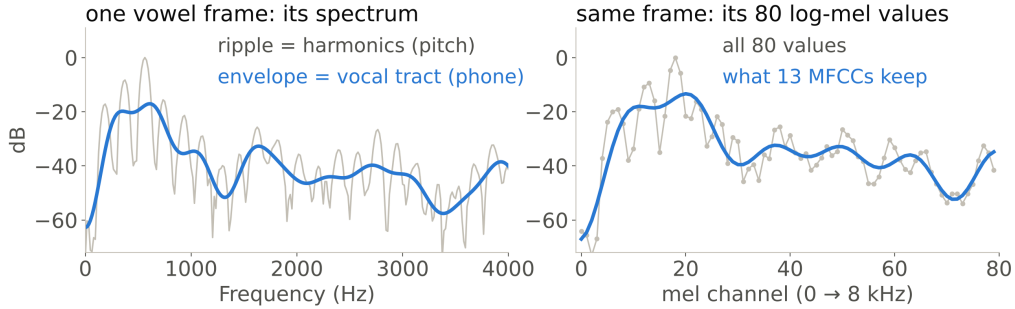
Part 5

MFCC

Intuition only (details: reading homework)

MFCC: Keep the Envelope, Drop the Pitch

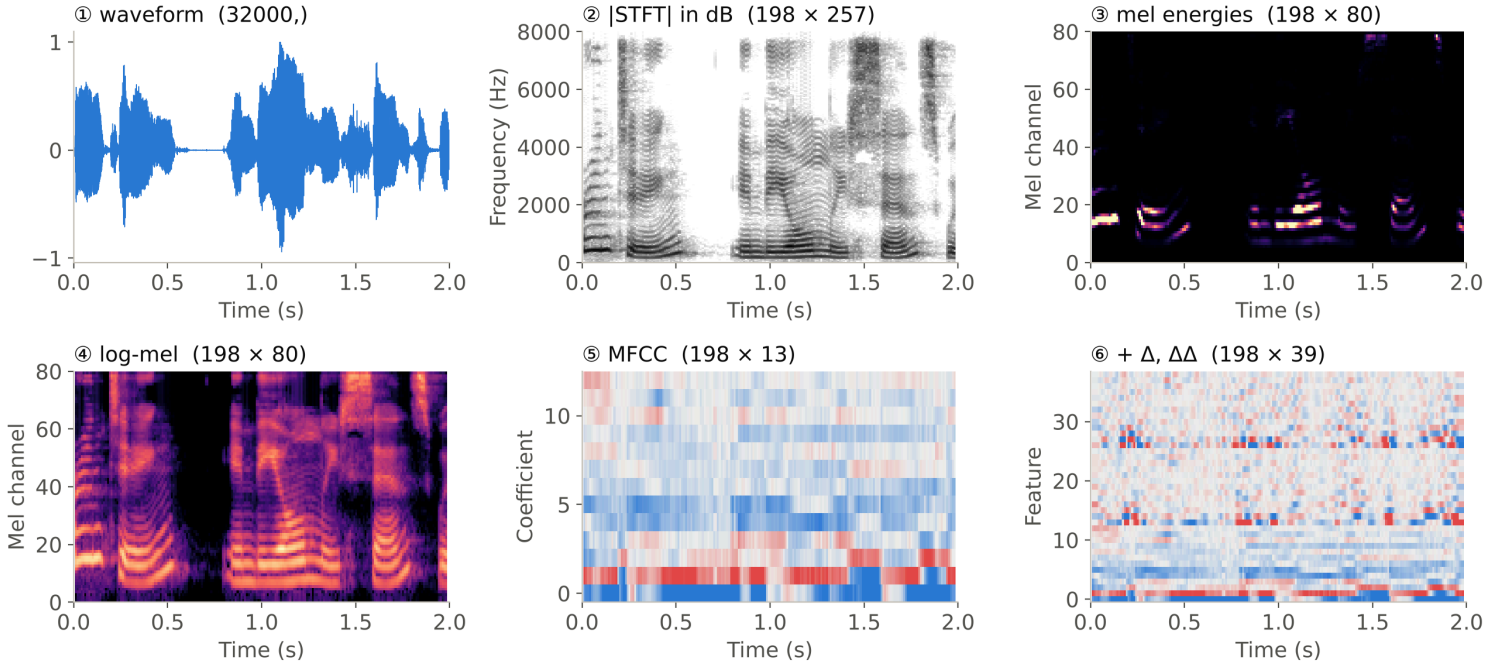
- **Source–filter** (Lecture 2): a vowel's spectrum is a fast **ripple** from the vocal folds (pitch) on a smooth **envelope** from the vocal tract. The phone is in the envelope.
- **MFCC** (mel-frequency cepstral coefficients) describe the *shape* of each log-mel frame with only **13 numbers**: enough for the envelope, too few for the ripple (blue, right).
- With how they change over time: **39 per frame**. Compact and much less correlated than log-mel, they suit the GMM-HMMs of next week; neural models mostly use log-mel.



Note
Details (cepstrum, DCT, deltas): reading homework, J&M Ch. 15.6.

Part 6
Conclude

The Whole Pipeline, One Utterance



198×400 frames → |DFT|² 198×257 → mel 198×80 → log → DCT 198×13 → +Δ,ΔΔ 198×39

Check Yourself

1. Put these steps in order: *log, mel filterbank, window, $|DFT|^2$, frame, DCT.*

2. A 16 kHz signal, 25 ms window, 10 ms hop, 512-point FFT. What are the bin spacing, the number of bins, and the number of frames in a 3 s clip?

3. Why do we window?

What to Take Away

- The **DFT** measures how much of each frequency a frame contains.
- We analyze short **frames** (L = 25 ms, H = 10 ms). Windowing with **Hamming/Hann** removes the edge jump that causes **spectral leakage**.
- The **STFT** is the DFT of every windowed frame; the **spectrogram** is its magnitude in dB.
- **Mel filterbank + log** gives log-mel: mel warps the frequency axis, the log (dB) compresses the energy axis. This is the standard neural input.
- **MFCC** keeps the vocal-tract envelope, drops most pitch detail, and decorrelates the features.

Next week: how do we model the *sequence* of these vectors? Alignment, HMMs, the forward algorithm, and Viterbi.

Assignment 3: Build the Front End

In the lab you will build the pipeline from this lecture.

1. **Frame and window** a recording yourself (NumPy only): check the frame count against the formula.
2. **STFT and spectrogram**: compute the DFT per frame and compare with `librosa.stft`. It returns bins × frames and pads by default (`center=False` and `n_fft=512`).
3. **Mel filterbank**: build the triangular filters and apply them, then take the log.
4. **MFCC**: keep 13 coefficients and add Δ and ΔΔ.
5. **Explore**: vary the window length and the window type, and describe what changes.

> **Tip**
> Reading: J&M Ch. 15.5.2–15.6 (windowing, DFT, mel filter bank and log, MFCC); review Ch. 15.4.5 from last week.

Learn More

Course Homepage · J&M Ch. 15 · librosa: melspectrogram · HuggingFace Audio Course · Chrome Music Lab: Spectrogram · But what is the Fourier Transform? A visual introduction